

Modern Developments in Sequential Testing and E-Processes

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Sequential Testing Problem

- Sequential testing arises from streaming data with adaptive decisions
 - ▶ Clinical trials, Online A/B testing
- Classical: Sequential Probability Ratio Test (Wald, 1945)
 - ▶ Simple hypothesis testing: $H_0 : X_i \stackrel{i.i.d.}{\sim} f_0, H_1 : X_i \stackrel{i.i.d.}{\sim} f_1$.

$$\Lambda_t = \prod_{i=1}^t \frac{f_1(X_i)}{f_0(X_i)} \begin{cases} > A & \text{reject } H_0 \\ < B & \text{accept } H_0 \\ \text{o.w.} & \text{continue sampling} \end{cases}$$

- ▶ Minimizes the expected sample size under fixed Type I and Type II error control.

E-value

- An E-value is a **nonnegative** random variable satisfying:

$$\sup_{\mathbb{P} \in H_0} \mathbb{E}_{\mathbb{P}}[E] \leq 1.$$

- ▶ E is interpreted as **evidence** against H_0 .
- ▶ Example: simple hypothesis testing $H_0 : X_i \stackrel{i.i.d.}{\sim} f_0$, $H_1 : X_i \stackrel{i.i.d.}{\sim} f_1$

$$E = \prod_{i=1}^n \frac{f_1(X_i)}{f_0(X_i)}.$$

- $\{E \geq 1/\alpha\}$ gives a valid hypothesis test at level α :

$$\mathbb{P}(E \geq \frac{1}{\alpha}) \leq \alpha \cdot \mathbb{E}_{\mathbb{P}}[E] \leq \alpha \quad \forall \mathbb{P} \in H_0.$$

E-values in Sequential Testing

- **A shift in perspective:**

Wald's SPRT (decision-theoretic) $\implies$ E-values (evidence-based)

- Simple hypothesis testing problem: $H_0 : X_i \stackrel{i.i.d.}{\sim} f_0$, $H_1 : X_i \stackrel{i.i.d.}{\sim} f_1$

$$E_{\textcolor{red}{t}} = \prod_{i=1}^t \frac{f_1(X_i)}{f_0(X_i)}.$$

- Reject H_0 whenever $E_t \geq 1/\alpha$: a valid sequential test at level α .
- **Ville's inequality:** If $(M_t)_{t \geq 0}$ is a nonnegative supermartingale and $M_0 \leq 1$,

$$\mathbb{P} \left(\sup_{t \geq 0} M_t \geq \frac{1}{\alpha} \right) \leq \alpha.$$

Testing supermartingales (SMs) and E-processes

- **Testing supermartingale (SM):** $(M_t)_{t \geq 0}$ is nonnegative,

$$M_0 \leq 1, \quad \mathbb{E}_{\mathbb{P}}[M_t | \mathcal{F}_{t-1}] \leq M_{t-1} \text{ for all } t \geq 1, \mathbb{P} \in H_0$$

However, a sequential test does **not necessarily** require a SM!

- **E-process:** $(E_t)_{t \geq 0}$ is nonnegative, and there exists a testing SM $(M_t)_{t \geq 0}$ such that $E_t \leq M_t$.
 - ▶ By Ville's inequality,

$$\mathbb{P} \left(\exists t \geq 0, E_t \geq \frac{1}{\alpha} \right) \leq \alpha.$$

- ▶ A nonnegative process $(E_t)_{t \geq 0}$ is an e-process if and only if^[1]

$$\sup_{\mathbb{P} \in H_0} \mathbb{E}_{\mathbb{P}}[E_{\tau}] \leq 1 \quad \forall \text{ stopping times } \tau.$$

Why Evidence-Based Sequential Testing?

- **Safe:** We may stop whenever the accumulated evidence is sufficiently large.
- **Anytime validity:** Evidence can be monitored continuously without invalidating Type I error control.
- **Sequential flexibility:** Sampling, stopping, and continuation rules may depend on the observed history.

Connection 1: Confidence Sequence

- Motivating problem: **Stochastic multi-armed bandit (MAB)**
Setting: K arms, each corresponding to a reward distribution.
Goal: With probability at least $1 - \delta$, select an optimal arm.
- **Naive approach:** Build a confidence interval for mean of each arm!
 - ▶ At the n th round: $\mu_1 \in [L_n^{(1)}, U_n^{(1)}], \dots, \mu_K \in [L_n^{(K)}, U_n^{(K)}]$.
 - ▶ Terminate when

$$L_n^{(i)} > \max_{j \neq i} U_n^{(j)}.$$

- Two problems:
 - ▶ Error inflation from comparing two confidence intervals
⇒ Bonferroni-type correction
 - ▶ Error inflation from repeatedly using **fixed-time** confidence intervals
⇒ **Problematic!**

Connection 1: Confidence Sequence

- A **confidence sequence** for an unknown quantity of interest θ is a sequence of confidence sets $(Cl_t)_{t=1}^{\infty}$,

$$\mathbb{P}(\forall t \geq 1 : \theta \in Cl_t) \geq 1 - \alpha.$$

- ▶ (P1) Nonasymptotic and nonparametric
- ▶ (P2) Unbounded sample size
- ▶ (P3) Arbitrary stopping rules
- ▶ (P4) Asymptotically zero width

- An E-process $(E_t(\theta))$ yields

$$Cl_t = \{\theta : E_t(\theta) \leq 1/\alpha\}.$$

- ▶ Example: $(X_t)_{t=1}^{\infty}$ from a 1-sub-Gaussian distribution with mean θ ^[2]

$$Cl_t = \frac{\sum_{i=1}^t X_i}{t} \pm 1.7 \sqrt{\frac{\log \log(2t) + 0.72 \log(10.4/\alpha)}{t}}.$$

Connection 2: Concentration inequality

- Classical: Hoeffding's inequality (Hoeffding, 1963)

If $X_i \in [-1, 1]$ for all i and at a **fixed time** m ,

$$\mathbb{P} \left(\sum_{i=1}^m X_i \geq x \right) \leq e^{-x^2/2m}.$$

- A **time-uniform** analogue of classical concentration inequalities^[3]:
Key observation: for $S_t = \sum_{i=1}^t X_i$ and V_t (variance process),

$$E_t = \exp\{\lambda S_t - \psi(\lambda) V_t\}$$

is an E-process.

$$\mathbb{P} \left(\exists t : S_t \geq \frac{\log(1/\alpha) + \psi(\lambda) V_t}{\lambda} \right) \leq \alpha.$$

Connection 2: Concentration inequality

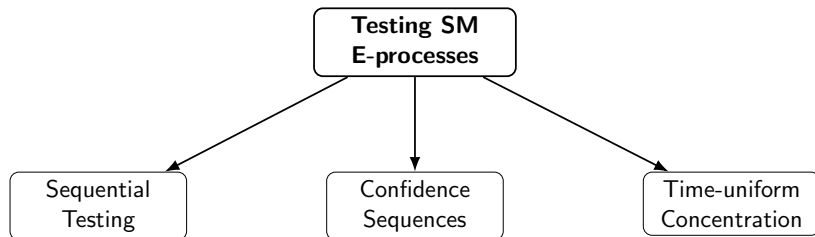
- Time-uniform Hoeffding inequality^[3]

If $X_t \in [-1, 1]$ for all t ,

$$\mathbb{P} \left(\exists t \in \mathbb{N} : S_t \geq x + \frac{x}{2m}(t - m) \right) \leq e^{-x^2/2m}.$$

- ▶ At $t = m$, this **recovers** Hoeffding's inequality.
- Many classical concentration inequalities were strengthened in a time-uniform form.^[3]
 - ▶ Bernstein (1927, moment constraint), Bennett (1962, bounded constraint), de la Peña (1999, self-normalized style), Tropp (2012, matrix-valued)

Conclusion



E-processes provide a common language for modern sequential inference!

References I

- [1] Aaditya Ramdas et al. *Admissible Anytime-Valid Sequential Inference Must Rely on Nonnegative Martingales*. 2020. [arXiv: 2009.03167 \[math.ST\]](#).
- [2] Steven R. Howard et al. “Time-uniform, Nonparametric, Nonasymptotic Confidence Sequences”. In: *The Annals of Statistics* 49.2 (2021), pp. 1055–1080. DOI: [10.1214/20-AOS1991](#).
- [3] Steven R. Howard et al. “Time-uniform Chernoff Bounds via Nonnegative Supermartingales”. In: *Probability Surveys* 17 (2020), pp. 257–317. DOI: [10.1214/18-PS321](#).